Mustafar Planetary Launch Simulation Design Report

Team Nexus

1 Introduction

When we first started to design this dual-stage rocket, we first decided to build the airframe entirely out of Cardboard (0.68 g/cm^3). It proved to be very helpful for maximizing flight apogee while maintaining relatively low weight.

Our mission was to reach the Apogee, deploy the parachutes while counteracting the wind component of 10 m/s and then safely recovering the payload. We had to fight against numerous variables, so we used the '**Trial and Error** ' tactic :D



Figure 1: Orbital rendering of the Team Nexus vehicle in flight

2 Rocket Architecture

To launch our rocket, we're using an inline staging approach. The lower half acts as our heavy booster (running an I300-14 motor). It gives us that massive kick off the rail so we don't get entirely blown over by crosswinds. Once it burns out, the coupler separates and the 2nd stage (G80-10) takes over to carry our payload the rest of the way up. We used classic clipped-delta fins on both stages to shift the Center of Pressure (CP) safely behind the Center of Gravity (CG).

One of the most critical parts of the design is the recovery system. We packed a dual-parachute setup beneath the payload module, separated by a bulk head. As shown in our designs, we separated their functions: **the first (upper) parachute is our Drogue Chute**. It pops out to slow down that initial, high-speed descent so our fins don't shear off. **The second (lower) parachute is the Main Chute**, which deploys for the soft touchdown.

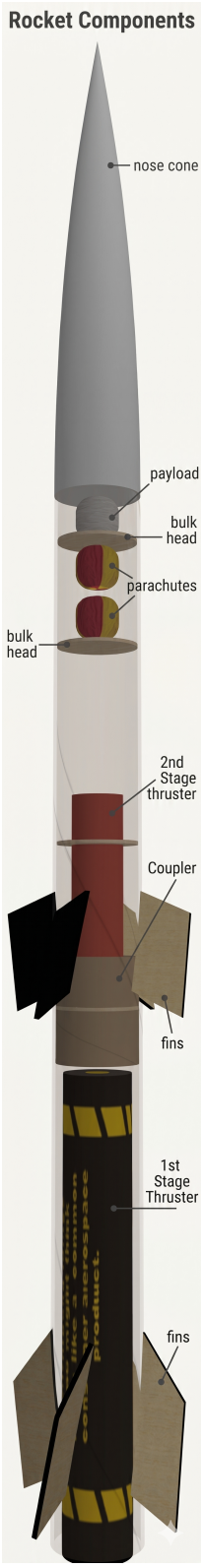


Figure 2: Labelled Diagram of our Rocket

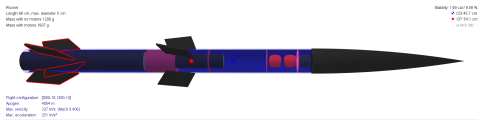


Figure 3: Profile View of our Rocket

Total Length	88.0 cm
Maximum Diameter	5.0 cm
Mass (Unloaded)	1290 g
Mass (Loaded with Motors)	1837 g
Motor Configuration	[1st Stage: I300-14] / [2nd Stage: G80-10]

Figure 4: Baseline Data point of Rocket

We faced a lot of problems with the horizontal component of velocity added due to the wind of 10 m/s, which led to very high velocities at Apogee does making it dangerous to deploy our parachutes, so to counter that we introduced Cant component to our fins which is set to 12°.

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3 Simulation on Planet Mustafar

The OpenRocket flight simulation was analysed to evaluate the rocket's trajectory, velocity, acceleration, stability, and recovery performance under the Planet Mustafar environment.

3.1 Flight Event Timeline

Table 1 summarizes the major events recorded during the simulation.

Table 1: Mustafar Flight Event Timeline

Time (s)	Flight Event
0.000	Motor ignition and launch event
0.020	Liftoff
0.093	Launch rod clearance
1.600	First-stage burnout
15.600	Ejection charge and stage separation
15.600	Second-stage ignition
17.097	Second-stage burnout
27.097	Ejection charge event
37.900	Apogee reached
37.901	Recovery deployment event
885.267	Ground impact and simulation end

3.2 Flight Performance Parameters

Table 2 presents the principal parameters extracted from the flight data.

Table 2: OpenRocket Simulation Parameters

Parameter	Value
Apogee	4063.91 m
Time to apogee	37.833 s
Maximum velocity	236.941 m/s
Maximum Mach number	0.406
Maximum total acceleration	251.095 m/s^2
Initial rocket mass	1836.877 g
Minimum rocket mass	1266.868 g
Lateral displacement at apogee	421.35 m
Lateral displacement at impact	8503.255 m
Vertical velocity at impact	−4.903 m/s
Total flight duration	885.267 s

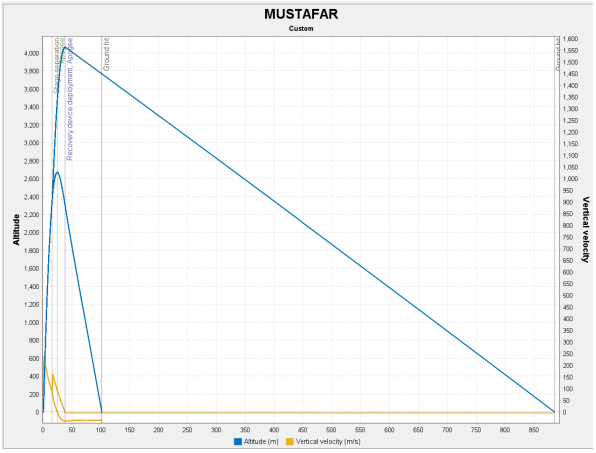


Figure 5: Flight data graph for the Mustafar simulation.

4 Earth–Mustafar Environmental Comparison

To evaluate the effect of the planetary environment, the same rocket design was simulated under Earth and Mustafar conditions. The flight data is used to compare apogee, velocity, Mach number, stability, and lateral drift. The results are summarized in Table 3.

4.1 Effect of Gravity

The Earth simulation used a gravitational acceleration of approximately 9.79 m/s^2, whereas the Mustafar simulation used 6.50 m/s^2. Lower gravity reduces the gravitational force acting on the rocket and can extend the coast phase. The recorded apogee increased from 1588.081 m on Earth to 4063.910 m on Mustafar. However, gravity was not varied independently, so this increase cannot be attributed to gravity alone.

4.2 Effect of Atmospheric Pressure and Density

The launch-site pressure decreased from approximately 1013.25 mbar on Earth to 750 mbar on Mustafar. The recorded launch density

Table 3: Comparison of Earth and Mustafar Simulations

Parameter	**Earth**	**Mustafar**
Apogee (m)	1588.081	4063.910
Maximum velocity (m/s)	220.571	236.941
Maximum Mach number	0.650	0.406
Maximum total acceleration (m/s^2)	1826.460	251.095
Lateral displacement at impact (m)	4897.250	8503.255

also decreased from approximately 1.0 kg/m^3 to 0.3733 kg/m^3.

Lower atmospheric density generally reduces aerodynamic drag during ascent, potentially allowing greater velocity and altitude. The maximum velocity increased from 220.571 m/s on Earth to 236.941 m/s on Mustafar. Since multiple environmental parameters changed simultaneously, the individual effects of pressure and density remain unverified.

Reduced density also affects parachute performance because less aerodynamic force is generated at a given descent speed. The Earth simulation reports high-speed deployment warnings for both recovery devices. Recovery safety and descent rates must therefore be assessed separately for each environment.

4.3 Effect of Temperature and Mach Number

The launch temperature was 15°C on Earth and 426.85°C on Mustafar, corresponding to the specified 700 K Mustafar surface temperature.

The recorded speed of sound increased from 340.389 m/s on Earth to 589.970 m/s on Mustafar. Despite its higher maximum velocity, Mustafar had a lower maximum Mach number: 0.406 compared with 0.650 on Earth. This is consistent with the higher recorded speed of sound. However, the maximum Mach number depends on the atmospheric conditions throughout the trajectory, not just at launch.

4.4 Effect of Wind

The mission specifies a 10 m/s wind on Mustafar. The simulations recorded lateral displacement at impact of 4897.250 m on Earth and 8503.255 m on Mustafar. The greater displacement is consistent with substantial wind-driven drift. However, lateral displacement also depends on flight duration, trajectory, the wind profile, and recovery descent. The difference cannot be attributed to wind alone without a controlled comparison.

4.5 Stability and Recovery

The Earth simulation reports high-speed deployment warnings for both parachutes, while the Mustafar simulation exhibits controlled descend as per the mission requirement of 4–8 m/s.

4.6 Conclusion of Environmental Comparison

The Mustafar simulation achieved an apogee of 4063.910 m, within the required range of 3900–4100 m. Its maximum velocity was higher than Earth's, while its maximum Mach number was lower. Lateral displacement at impact was also greater under the Mustafar simulation conditions.

These results are consistent with the environmental hypotheses, but they do not isolate the effect of each parameter because gravity, pressure, temperature, density, and wind changed together. Further simulations that vary one parameter at a time are required. Stability, parachute deployment, and component-specific descent rates must also be verified before concluding that all mission requirements have been satisfied.

5 Conclusion

The OpenRocket simulation evaluated the performance of the Team Nexus dual-stage rocket under the environmental conditions of Planet Mustafar. The rocket achieved a maximum altitude of 4063.91 m in 37.833 s, satisfying the mission's target apogee range of 3900–4100 m. The recorded maximum velocity was 236.941 m/s, with a maximum Mach number of 0.406.

The design process also highlighted the importance of stability, wind compensation, and recovery-system performance. Although the simulation reached the target apogee, the rocket's lateral drift and parachute deployment behaviour remain important considerations when evaluating overall mission safety.

In conclusion, the simulation demonstrates that the Team Nexus rocket can achieve the required apogee under the specified Mustafar conditions.